

CprE 3810 – RISC-V Single-Cycle Processor: Complete Control Signals Table

Instruction Encoding					Core Control Signals					Datapath Routing					Special / Memory			
Inst	Type	Opcode [6:0]	Funct3 [14:12]	Funct7/Imm [31:25]	s_RegWrite (RegWr)	s_MemWrite (DMemWr)	s_ALUSrc 0=rs2 1=imm B_SRC_MUX	s_IsAUIPC (ALUSrcA) 0=rs1 1=PC AUIPC_MUX	s_ALUCtrl (4-bit)	ImmType (3-bit)	s_MemToReg (ResultSrc 2-bit)	s_PC_SRC (PCSrc 2-bit)	s_Branch	s_MemRead	s_LoadData funct3 (LoadType)	s_IsLUI (LUIsel)	s_Halt	
add	R	0110011	000	0000000	1	0	0	0	0000	---	00	00	0	0	---	0	0	
sub	R	0110011	000	0100000	1	0	0	0	0001	---	00	00	0	0	---	0	0	
and	R	0110011	111	0000000	1	0	0	0	0010	---	00	00	0	0	---	0	0	
or	R	0110011	110	0000000	1	0	0	0	0011	---	00	00	0	0	---	0	0	
xor	R	0110011	100	0000000	1	0	0	0	0100	---	00	00	0	0	---	0	0	
slt	R	0110011	010	0000000	1	0	0	0	0101	---	00	00	0	0	---	0	0	
sll	R	0110011	001	0000000	1	0	0	0	0111	---	00	00	0	0	---	0	0	
srl	R	0110011	101	0000000	1	0	0	0	1000	---	00	00	0	0	---	0	0	
sra	R	0110011	101	0100000	1	0	0	0	1001	---	00	00	0	0	---	0	0	
addi	I	0010011	000	----	1	0	1	0	0000	000	00	00	0	0	---	0	0	
andi	I	0010011	111	----	1	0	1	0	0010	000	00	00	0	0	---	0	0	
ori	I	0010011	110	----	1	0	1	0	0011	000	00	00	0	0	---	0	0	
xori	I	0010011	100	----	1	0	1	0	0100	000	00	00	0	0	---	0	0	
slti	I	0010011	010	----	1	0	1	0	0101	000	00	00	0	0	---	0	0	
sltiu	I	0010011	011	----	1	0	1	0	0110	000	00	00	0	0	---	0	0	
slli	I	0010011	001	0000000	1	0	1	0	0111	000	00	00	0	0	---	0	0	
srlr	I	0010011	101	0000000	1	0	1	0	1000	000	00	00	0	0	---	0	0	
srair	I	0010011	101	0100000	1	0	1	0	1001	000	00	00	0	0	---	0	0	
lw	I	0000011	010	----	1	0	1	0	0000	000	01	00	0	1	010	0	0	
lh	I	0000011	001	----	1	0	1	0	0000	000	01	00	0	1	001	0	0	
lhu	I	0000011	101	----	1	0	1	0	0000	000	01	00	0	1	101	0	0	
lb	I	0000011	000	----	1	0	1	0	0000	000	01	00	0	1	000	0	0	
lbu	I	0000011	100	----	1	0	1	0	0000	000	01	00	0	1	100	0	0	
sw	S	0100011	010	----	0	1	1	0	0000	001	00	00	0	0	---	0	0	
beq	B	1100011	000	----	0	0	0	0	0001	010	00	01	1	0	---	0	0	
bne	B	1100011	001	----	0	0	0	0	0001	010	00	01	1	0	---	0	0	
blt	B	1100011	100	----	0	0	0	0	0001	010	00	01	1	0	---	0	0	
bge	B	1100011	101	----	0	0	0	0	0001	010	00	01	1	0	---	0	0	
bltu	B	1100011	110	----	0	0	0	0	0001	010	00	01	1	0	---	0	0	
bgeu	B	1100011	111	----	0	0	0	0	0001	010	00	01	1	0	---	0	0	
jal	J	1101111	---	----	1	0	0	0	----	100	10	01	0	0	---	0	0	
jalr	I	1100111	000	----	1	0	1	0	0000	000	10	10	0	0	---	0	0	
lui	U	0110111	---	----	1	0	1	0	1010	011	00	00	0	0	---	1	0	
auiipc	U	0010111	---	----	1	0	1	1	0000	011	00	00	0	0	---	0	0	
wfi	I	1110011	000	000100000101	0	0	0	0	----	000	00	00	0	0	---	0	1	